

SLASH MARK IT SOLUTIONS

Business Analyst Internship Project Report

BANK CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

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CERTIFICATE

This is to certify that Ms. Kavya P, a student of Bachelor of Engineering in Computer Science and Business Systems (CSBS) from St. Joseph Engineering College, Mangaluru, has successfully completed the Business Analyst Internship at Slash Mark IT Solutions Pvt. Ltd.

During the internship, she successfully carried out the project titled “Bank Customer Churn Prediction Using Machine Learning.”

The objective of the project was to analyze customer data and develop machine learning models to predict customer churn. Various data preprocessing, exploratory data analysis (EDA), feature engineering, model training, hyperparameter tuning, and model evaluation techniques were implemented to identify customers who are likely to discontinue banking services. The project also focused on understanding business risks, identifying customer retention opportunities, and providing data-driven recommendations to improve customer satisfaction and reduce churn.

Throughout the internship, she demonstrated dedication, analytical thinking, problem-solving ability, and a strong willingness to learn new technologies and business analytics concepts. The project was completed successfully within the stipulated internship duration and met the objectives assigned under the Business Analyst Internship program.

We appreciate her sincere efforts and wish her success in all her future academic and professional endeavors.

Authorized Signatory

Slash Mark IT Solutions Pvt. Ltd.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Slash Mark IT Solutions Pvt. Ltd. for providing me with the opportunity to undertake the Business Analyst Internship and work on the project titled “Bank Customer Churn Prediction Using Machine Learning.” This internship has been an enriching learning experience that enhanced my understanding of business analytics, machine learning, and data-driven decision-making.

I am also grateful to my family and friends for their continuous encouragement and motivation throughout the successful completion of this project. Completing this internship has strengthened my analytical thinking, problem-solving abilities, and understanding of how business analytics can be applied to solve organizational challenges.

EXECUTIVE SUMMARY

Customer churn is one of the most critical challenges faced by financial institutions and banking organizations. Acquiring new customers is significantly more expensive than retaining existing ones, making customer retention a strategic priority for sustainable business growth. Identifying customers who are likely to discontinue banking services enables organizations to take proactive measures, improve customer satisfaction, and minimize revenue loss. Therefore, predictive analytics has become an essential component of modern customer relationship management.

This project focuses on Bank Customer Churn Prediction Using Machine Learning with the objective of identifying customers who are at a high risk of leaving the bank and recommending effective customer retention strategies. The project integrates Business Analytics, Exploratory Data Analysis (EDA), Feature Engineering, and Machine Learning techniques to transform raw customer data into actionable business insights. By analyzing customer demographics, financial information, account activity, and banking behavior, the project aims to understand the major factors influencing customer churn and develop predictive models that assist decision-makers in reducing customer attrition.

The analysis began with comprehensive data preprocessing, including data inspection, missing value verification, removal of irrelevant features, feature engineering, categorical variable encoding, feature scaling, and train-test data splitting. Exploratory Data Analysis (EDA) was then performed to study customer characteristics, identify behavioral patterns, and examine the relationships between various customer attributes and churn. Statistical visualizations and business-oriented interpretations were used to gain meaningful insights into customer behavior.

To identify the most suitable predictive model, multiple machine learning algorithms were developed and evaluated, including Logistic Regression, Polynomial Logistic Regression, Support Vector Machine (RBF Kernel), Support Vector Machine (Polynomial Kernel), Random Forest, and XGBoost. Model performance was assessed using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, ROC Curve, and Area Under the Curve (AUC). Hyperparameter tuning techniques, including Grid Search, were employed to improve model performance by identifying the optimal parameter combinations. The project also discusses the significance of SHAP (SHapley Additive exPlanations) as an explainable artificial intelligence technique for interpreting model predictions and enhancing stakeholder confidence in machine learning solutions.

The findings indicate that customer churn is influenced by several business-related factors, including customer age, account balance, tenure, geographical location, activity status, number of banking products, and estimated salary. The comparative analysis of multiple machine learning models helped identify the most effective algorithm for predicting customer churn while maintaining a balance between predictive performance and business interpretability.

Based on these findings, practical business recommendations have been proposed to support customer retention, including personalized engagement strategies, proactive intervention for high-risk customers, loyalty programs, customer segmentation, predictive monitoring systems, and data-driven decision-making. The project demonstrates how Business Analytics and Machine Learning can assist banking institutions in reducing customer attrition, improving customer lifetime value, strengthening customer relationships, and achieving long-term business sustainability.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's highly competitive banking industry, retaining existing customers has become equally as important as acquiring new customers. Customers have access to a wide range of financial institutions offering similar products and services, making it easier for them to switch to competitors when their expectations are not met. Customer churn, which refers to the loss of customers over a period of time, has become a significant business challenge that directly impacts an organization's profitability, customer lifetime value, and market competitiveness.

Banks invest substantial financial and operational resources in acquiring new customers through marketing campaigns, promotional offers, and customer acquisition programs. However, studies have consistently shown that retaining an existing customer is considerably more cost-effective than acquiring a new one. Therefore, understanding the reasons behind customer churn and identifying customers who are at a high risk of leaving have become critical priorities for modern banking organizations.

With the rapid advancement of Business Analytics, Artificial Intelligence, and Machine Learning, organizations are increasingly leveraging predictive analytics to support strategic decision-making. Predictive models enable businesses to analyze historical customer data, identify hidden behavioral patterns, and forecast future customer actions. These insights allow organizations to implement proactive customer retention strategies before customers decide to discontinue their services.

This internship project focuses on Bank Customer Churn Prediction Using Machine Learning, where historical customer information is analyzed to predict whether a customer is likely to exit the bank. The project combines data preprocessing, exploratory data analysis (EDA), feature engineering, predictive modeling, and business interpretation to develop an effective customer churn prediction system. Multiple machine learning algorithms, including Logistic Regression, Support Vector Machine (SVM), Random Forest, and Extreme Gradient Boosting (XGBoost), are implemented and evaluated to determine the most suitable predictive model.

Apart from developing predictive models, this project emphasizes the importance of Business Analytics in supporting organizational decision-making. The insights generated from the analysis help identify business risks associated with customer attrition and provide practical customer retention strategies that can improve customer satisfaction, strengthen customer loyalty, and enhance long-term organizational performance.

Overall, this project demonstrates how machine learning techniques can be integrated with business analytics to transform raw customer data into meaningful insights, enabling banking institutions to make informed, data-driven decisions while improving customer retention and operational efficiency.

1.2 Background

Customer churn is one of the most significant challenges faced by banks and other financial service providers. Churn occurs when customers voluntarily discontinue their relationship with a financial institution and move to competing organizations. Such customer loss affects not only immediate revenue but also long-term business growth, customer acquisition costs, and brand reputation.

Several factors influence customer churn, including customer satisfaction, service quality, product offerings, pricing strategies, digital banking experience, relationship management, and overall customer engagement. While some of these factors are observable through customer feedback, many behavioral patterns remain hidden within organizational data. Business Analytics enables organizations to uncover these hidden relationships and convert data into actionable business knowledge.

Machine Learning has emerged as one of the most effective techniques for predicting customer behavior. By learning from historical customer data, machine learning algorithms can identify complex patterns that are difficult to detect using traditional statistical methods. These predictive capabilities enable banks to identify high-risk customers, prioritize retention efforts, and optimize marketing strategies based on customer behavior.

The increasing availability of structured customer data, combined with advancements in computational techniques, has encouraged financial institutions to adopt predictive analytics for customer relationship management. Organizations can now develop intelligent systems capable of continuously monitoring customer behavior, estimating churn probability, and recommending personalized retention strategies.

This project applies these analytical concepts using a publicly available banking dataset and demonstrates how Business Analytics and Machine Learning can work together to solve real-world customer retention problems.

1.3 Customer Churn in Banking

Customer churn refers to the percentage of customers who discontinue using a company's products or services within a given period. In the banking sector, customer churn may occur when customers close their accounts, transfer their savings to another bank, stop using banking products, or completely terminate their relationship with the institution.

The banking industry experiences customer churn due to multiple business and customer-related factors, including:

- Poor customer service and support.
- Better financial products offered by competitors.
- High service charges and transaction fees.
- Limited digital banking facilities.
- Lack of personalized customer engagement.
- Low customer satisfaction and trust.
- Ineffective complaint resolution.
- Changing financial needs and customer preferences.

Customer churn has serious business implications, including reduced revenue, increased customer acquisition costs, loss of cross-selling opportunities, and declining customer lifetime value. Therefore, predicting customer churn has become an important application of Business Analytics and Machine Learning.

By accurately identifying customers who are likely to leave, banks can implement targeted retention programs such as personalized offers, loyalty rewards, financial advisory services, and proactive

customer engagement. These strategies not only reduce customer attrition but also improve overall customer experience and organizational profitability.

The successful implementation of customer churn prediction systems enables financial institutions to shift from reactive decision-making to proactive customer relationship management, ensuring sustainable business growth in an increasingly competitive marketplace.

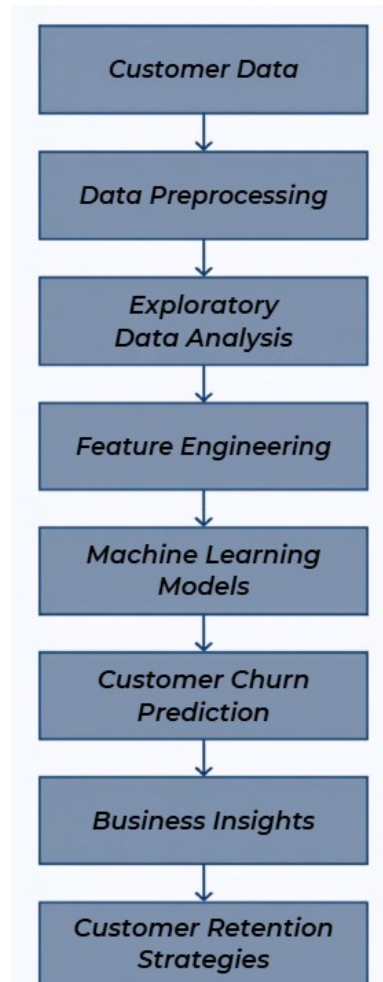


Figure 1.1: Overview of Customer Churn Prediction Process

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

Customer retention has become one of the most significant challenges in the banking industry due to increasing market competition, evolving customer expectations, and the availability of diverse financial service providers. Every customer who leaves the bank not only results in an immediate loss of revenue but also increases customer acquisition costs, reduces customer lifetime value, and affects long-term organizational profitability. Traditional methods of identifying customers who are likely to leave are often reactive, relying on customer complaints or account closures after the decision to leave has already been made.

Banks generate large volumes of customer data through daily transactions, account activities, product usage, and customer interactions. However, much of this valuable information remains underutilized when making strategic business decisions. By leveraging Business Analytics and Machine Learning, organizations can transform historical customer data into predictive insights that enable proactive customer retention.

The primary challenge addressed in this project is the development of a predictive model capable of accurately identifying customers who are at risk of churn. In addition to prediction, the project focuses on understanding the business factors that contribute to customer attrition and identifying practical retention strategies that can help improve customer satisfaction and reduce customer loss.

Furthermore, the project explores model optimization techniques such as Grid Search for hyperparameter tuning and discusses the role of SHAP (SHapley Additive exPlanations) in improving model transparency and explainability. These techniques support the development of reliable, interpretable, and business-oriented predictive solutions that can assist banking institutions in making informed decisions.

2.2 Objectives

The primary objective of this project is to develop an effective machine learning-based customer churn prediction system that enables banking organizations to identify customers who are likely to discontinue their relationship with the bank and implement timely customer retention strategies.

The specific objectives of this project are:

- To analyze customer demographic, financial, and behavioral data for churn prediction.
- To perform data preprocessing, including data inspection, feature engineering, encoding, and feature scaling.
- To conduct Exploratory Data Analysis (EDA) to understand customer behavior and identify patterns associated with churn.
- To build multiple machine learning models, including Logistic Regression, Polynomial Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost.
- To optimize model performance using Grid Search for hyperparameter tuning.
- To compare different machine learning algorithms using performance evaluation metrics such as Accuracy, Precision, Recall, F1-Score, ROC Curve, and AUC Score.
- To understand the business factors that influence customer churn.

- To recommend practical customer retention strategies based on analytical findings.
- To discuss the importance of Explainable Artificial Intelligence (SHAP) in improving model transparency and stakeholder trust.

2.3 Scope of the Project

The scope of this project is limited to the analysis of a structured customer churn dataset obtained from a publicly available banking dataset. The project focuses on applying Business Analytics and Machine Learning techniques to predict customer churn using historical customer information.

The project includes data preprocessing, exploratory data analysis, feature engineering, model development, hyperparameter tuning, model evaluation, and business interpretation of the analytical results. Multiple supervised machine learning algorithms are implemented and compared to determine the most suitable predictive model for customer churn prediction.

The project does not involve deployment of the predictive model into a production banking environment or integration with live banking systems. Instead, it focuses on demonstrating how predictive analytics can support strategic decision-making and improve customer retention through data-driven insights.

Although the project is based on historical data, the analytical framework and methodology can be extended to real-time banking applications, customer relationship management (CRM) systems, and enterprise-level predictive analytics solutions.

2.4 Business Significance

Customer churn has a direct impact on the financial performance and long-term sustainability of banking institutions. Retaining existing customers is generally more cost-effective than acquiring new customers, making customer retention a strategic business priority.

The implementation of predictive analytics enables banks to shift from reactive customer management to proactive customer engagement. By identifying customers who are likely to leave before they actually discontinue banking services, organizations can design targeted intervention strategies that improve customer satisfaction and strengthen customer loyalty.

The insights generated through this project support various business functions, including customer relationship management, marketing strategy, risk management, product development, and strategic decision-making. The predictive models developed during this project assist management in identifying high-risk customer segments, allocating retention resources efficiently, and improving overall operational performance.

Additionally, the application of machine learning techniques such as Random Forest and XGBoost demonstrates the growing role of Artificial Intelligence in Business Analytics. The discussion on SHAP explainability highlights the importance of building transparent and trustworthy AI systems that enable business stakeholders to understand and confidently utilize predictive models in organizational decision-making.

Overall, this project demonstrates how Business Analytics, Machine Learning, and Explainable Artificial Intelligence can work together to improve customer retention, reduce business risk, and enhance organizational competitiveness in the banking sector.

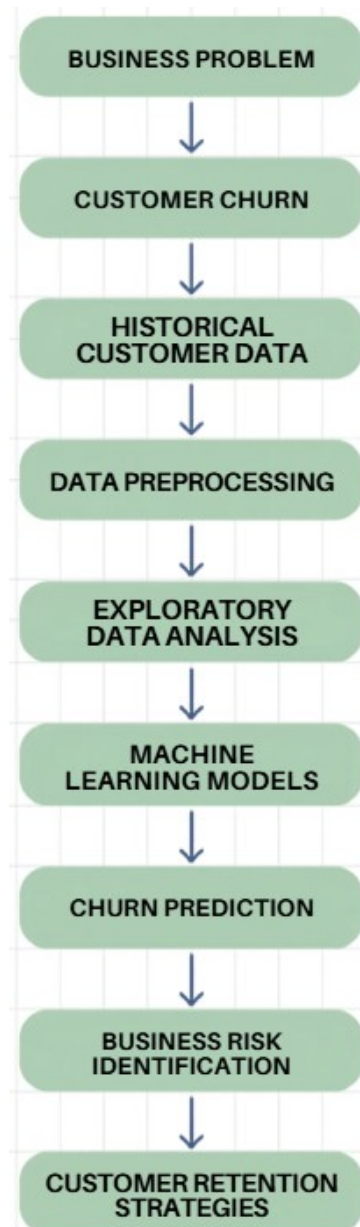


Figure 2.1: Business Problem to Business Solution Framework

CHAPTER 3: DATASET DESCRIPTION AND PROJECT METHODOLOGY

The success of any Business Analytics or Machine Learning project largely depends on the quality of the dataset and the methodology adopted for analysis. A well-structured dataset provides meaningful information about customer behavior, while a systematic analytical methodology ensures that the collected data is transformed into actionable business insights.

This chapter presents a detailed overview of the customer churn dataset used in this project, describes the important customer attributes, and explains the complete analytical workflow followed to predict customer churn. The methodology combines Business Analytics, Exploratory Data Analysis (EDA), Feature Engineering, Machine Learning, and Business Interpretation to develop an effective customer churn prediction model.

3.1 Dataset Description

The project utilizes the Bank Customer Churn Dataset, a publicly available dataset that contains customer demographic information, banking relationship details, financial characteristics, and customer activity records. The dataset is widely used for customer churn prediction research and provides sufficient information for understanding customer behavior and building predictive machine learning models.

The dataset consists of 1,000 customer records with 14 attributes, where each record represents an individual customer and their relationship with the bank. The target variable in the dataset is Exited, which indicates whether the customer has left the bank.

The dataset contains a combination of numerical and categorical variables, allowing multiple machine learning algorithms to identify complex relationships between customer characteristics and churn behavior.

Dataset Overview

Attribute	Description
Number of Records	1,000
Number of Features	14
Target Variable	Exited
Prediction Type	Binary Classification
Domain	Banking
Dataset Type	Structured Tabular Data

3.2 Description of Features

The dataset contains customer demographic, financial, and account-related information that may influence customer churn.

Feature	Description
RowNumber	Unique row identifier (removed during preprocessing).
CustomerId	Unique customer identification number (removed during preprocessing).
Surname	Customer's surname (removed during preprocessing).
CreditScore	Customer's creditworthiness score.
Geography	Country where the customer resides.
Gender	Customer gender.
Age	Age of the customer.
Tenure	Number of years the customer has been associated with the bank.
Balance	Account balance maintained by the customer.
NumOfProducts	Number of banking products used.
HasCrCard	Indicates whether the customer owns a credit card.
IsActiveMember	Indicates whether the customer actively uses banking services.
EstimatedSalary	Estimated annual salary of the customer.
Exited	Target variable indicating whether the customer left the bank (1 = Yes, 0 = No).

3.3 Project Methodology

The project follows a structured Business Analytics methodology to convert raw customer information into meaningful business insights and predictive models. The complete workflow adopted in this project is illustrated in Figure 3.1 and described in the steps below.

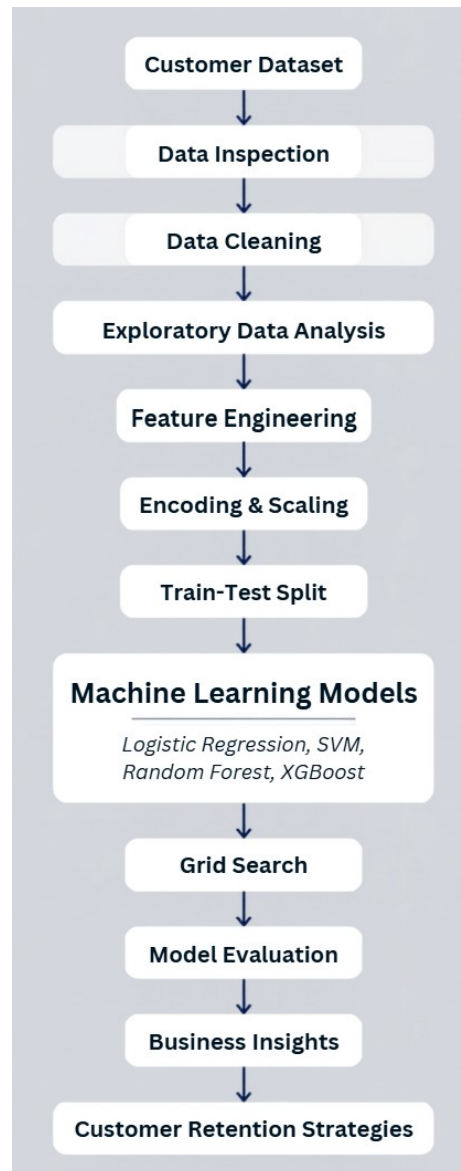


Figure 3.1: Project Methodology Workflow

Step 1 – Data Collection

The Bank Customer Churn dataset was imported into Python using the Pandas library. The dataset was inspected to understand its structure, dimensions, feature types, and target variable.

Step 2 – Data Understanding

The dataset was explored using descriptive statistics and structural analysis. Missing values, unique values, data types, and feature distributions were examined to assess overall data quality.

Step 3 – Data Preprocessing

Irrelevant variables such as RowNumber, CustomerId, and Surname were removed because they do not contribute to customer churn prediction. The dataset was checked for missing values, duplicate records, and inconsistencies before proceeding with analysis.

Step 4 – Exploratory Data Analysis (EDA)

EDA was performed using various statistical visualizations, including pie charts, count plots, box plots, and correlation analysis. This stage helped identify customer behavior patterns, churn distribution, demographic trends, and relationships between different customer characteristics.

Step 5 – Feature Engineering

To improve predictive performance, three additional features were created: BalanceSalaryRatio, TenureByAge, and CreditScoreGivenAge. These engineered features provide deeper insights into customer financial behavior and banking relationships.

Step 6 – Data Transformation

Categorical variables such as Geography and Gender were converted into numerical representations using One-Hot Encoding. Numerical variables were standardized using Min-Max Scaling to ensure that all features contributed equally during model training.

Step 7 – Train-Test Split

The processed dataset was divided into training and testing datasets using an 80:20 ratio. The training dataset was used to build machine learning models, while the testing dataset was reserved for evaluating model performance.

Step 8 – Machine Learning Model Development

Multiple supervised machine learning algorithms were implemented, including Logistic Regression, Polynomial Logistic Regression, Support Vector Machine (RBF Kernel), Support Vector Machine (Polynomial Kernel), Random Forest, and XGBoost. Each algorithm was trained independently and evaluated using common performance metrics.

Step 9 – Hyperparameter Tuning

To improve model performance, Grid Search Cross Validation (GridSearchCV) was applied for parameter optimization. Hyperparameter tuning helped identify the best parameter combinations for each machine learning model.

Step 10 – Model Evaluation

The trained models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC Curve, and Area Under Curve (AUC). The evaluation process helped determine the most effective predictive model for customer churn.

Step 11 – Business Interpretation

The final stage involved interpreting analytical findings from a business perspective. Instead of focusing solely on prediction accuracy, the project emphasized identifying customer churn patterns, business risks, and practical customer retention strategies that banking organizations can implement to improve customer loyalty and reduce attrition.

3.4 Software and Tools Used

The project was developed using Python-based data analytics and machine learning libraries. These tools were selected for their efficiency, scalability, and extensive support for predictive analytics.

Software / Tool	Purpose
Python	Programming language for data analysis and machine learning

Software / Tool	Purpose
Jupyter Notebook	Interactive development environment
Pandas	Data loading and manipulation
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Machine learning algorithms and preprocessing
XGBoost	Gradient boosting algorithm for predictive modeling
GridSearchCV	Hyperparameter tuning
GitHub	Project version control and submission

CHAPTER 4: DATA PREPROCESSING

4.1 Introduction

Data preprocessing is one of the most critical stages in any Business Analytics and Machine Learning project. Raw datasets often contain inconsistencies, irrelevant attributes, missing values, duplicate records, and features that may negatively influence model performance. Therefore, preprocessing ensures that the dataset is clean, consistent, and suitable for analysis.

In this project, several preprocessing techniques were applied to improve the quality of the banking customer dataset before developing predictive models. These techniques included dataset inspection, missing value analysis, removal of irrelevant attributes, feature engineering, categorical variable encoding, feature scaling, and train-test data splitting. Each preprocessing step contributed to improving the efficiency, reliability, and predictive capability of the machine learning models.

4.2 Dataset Inspection

The dataset was initially imported into the Python environment using the Pandas library. Preliminary inspection was carried out to understand the overall structure of the dataset, including the number of observations, available features, and data types.

Functions such as `head()`, `shape`, `info()`, and `describe()` were used to examine the dataset and verify that all attributes were loaded correctly. The dataset contained 1,000 customer records and 14 features, representing customer demographic information, banking relationship details, financial characteristics, and the target variable (Exited).

Observation

The dataset was successfully loaded without any structural issues. It consisted of both numerical and categorical variables suitable for predictive modeling.

Business Interpretation

Understanding the structure of the dataset helps analysts identify the type of information available and determine the preprocessing techniques required before performing machine learning analysis.

4.3 Data Cleaning

Data cleaning ensures that the dataset used for model training is accurate and free of redundant information. Two checks were carried out at this stage: verifying that no customer record was duplicated, and removing attributes that serve only as identifiers rather than genuine predictive features.

4.3.1 Duplicate Record Verification

Duplicate records can introduce bias by giving disproportionate importance to repeated observations. Therefore, the dataset was examined to ensure that each customer record was unique. The inspection confirmed that duplicate records were not present within the dataset.

Observation

No duplicate customer records were identified.

Business Interpretation

The absence of duplicate entries improves data reliability and ensures that predictive models learn genuine customer behavior rather than repeated information.

4.3.2 Removal of Irrelevant Features

Not every variable contributes to predicting customer churn. Certain attributes simply act as unique identifiers and do not provide meaningful information for machine learning. The following features were removed during preprocessing:

Removed Feature	Reason for Removal
RowNumber	Serves only as a sequential record identifier
CustomerId	Unique customer identifier with no predictive value
Surname	Personal identifier unrelated to churn prediction

Removing these variables reduced unnecessary complexity and improved model efficiency.

Observation

Three identifier-based features were excluded from further analysis.

Business Interpretation

Eliminating irrelevant variables enables machine learning algorithms to focus on meaningful customer characteristics that influence churn.

4.4 Missing Value Analysis

Missing values reduce data quality and may negatively affect machine learning performance. Therefore, the dataset was examined to determine whether any attributes contained null or missing observations. The missing value analysis confirmed that all variables contained complete information, and no missing records were identified.

Observation

The dataset contains no missing values, indicating that all customer records are complete.

Business Interpretation

Since no missing values were detected, the dataset was immediately suitable for further preprocessing and model development without requiring imputation or deletion techniques.

4.5 Feature Engineering

Feature engineering was performed to derive additional variables capable of capturing customer behavior more effectively than the original dataset. Three new features were created:

Engineered Feature	Formula	Business Purpose
BalanceSalaryRatio	$\text{Balance} \div \text{EstimatedSalary}$	Indicates the proportion of account balance relative to customer income
TenureByAge	$\text{Tenure} \div \text{Age}$	Represents customer loyalty relative to age
CreditScoreGivenAge	$\text{CreditScore} \div \text{Age}$	Measures creditworthiness adjusted for customer age

These engineered variables provide additional information that may improve the predictive capability of machine learning algorithms.

Observation

Three derived variables were successfully generated from existing customer attributes.

Business Interpretation

Feature engineering allows the model to identify hidden relationships between customer financial behavior and churn that may not be captured by the original variables.

4.6 Data Encoding

Machine learning algorithms require numerical inputs for model training. Therefore, categorical variables such as Geography and Gender were transformed into numerical representations using One-Hot Encoding. This preprocessing step ensures that categorical information is represented without introducing artificial numerical relationships.

Observation

Categorical variables were successfully converted into binary numerical features.

Business Interpretation

Encoding enables predictive models to utilize customer demographic information while preserving the independence of different categories.

4.7 Feature Scaling

The dataset contains variables measured on different numerical scales. For example, Credit Score ranges from hundreds, while Balance and Estimated Salary are measured in much larger values. Without scaling, variables with larger magnitudes may dominate the learning process. To address this issue, Min-Max Scaling was applied, transforming numerical variables into a common range between 0 and 1.

Observation

All numerical variables were normalized using Min-Max Scaling before model training.

Business Interpretation

Feature scaling improves model convergence, enhances prediction performance, and ensures that all variables contribute proportionately during learning.

4.8 Train-Test Split

The processed dataset was divided into training and testing datasets using an 80:20 ratio.

- Training Dataset (80%): Used to train the machine learning models.
- Testing Dataset (20%): Used to evaluate model performance on previously unseen customer data.

This separation helps measure the ability of the models to generalize beyond the training data.

Observation

The dataset was successfully partitioned into training and testing subsets.

Business Interpretation

Separating training and testing data provides a realistic assessment of model performance and reduces the risk of overfitting.

4.9 Summary

This chapter described the preprocessing techniques applied to prepare the customer churn dataset for predictive modeling. The dataset was inspected, verified for missing values and duplicate records, cleaned by removing irrelevant features, enriched through feature engineering, transformed using one-hot encoding, normalized using Min-Max Scaling, and divided into training and testing datasets. These preprocessing steps established a reliable foundation for Exploratory Data Analysis and subsequent machine learning model development.

CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

Exploratory Data Analysis (EDA) is an important stage in Business Analytics that helps analysts understand the characteristics, distribution, and relationships within a dataset before developing predictive models. It enables the identification of trends, anomalies, and hidden patterns that may influence business decisions.

In this project, EDA was performed to examine customer demographics, banking behavior, financial characteristics, and their relationship with customer churn. Various statistical visualizations, including pie charts, count plots, box plots, histograms, and correlation analysis, were used to identify important patterns within the dataset. The insights obtained from this analysis guided feature engineering and supported the development of machine learning models for churn prediction.

5.2 Target Variable Analysis

The first step in EDA involved analyzing the target variable *Exited*, which indicates whether a customer has left the bank.

Proportion of customer churned and retained

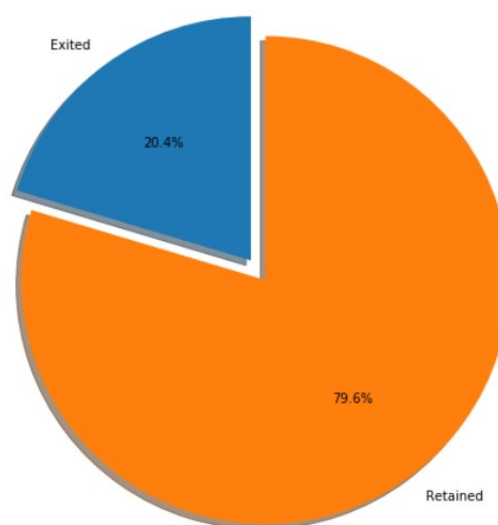


Figure 5.1: Customer Churn Distribution (Pie Chart)

Observation

The pie chart shows that 79.6% of customers were retained while 20.4% churned — a ratio of roughly 4 to 1 in favor of retained customers.

Business Interpretation

The dataset is imbalanced, with retained customers significantly outnumbering churned customers. This indicates that churn is relatively uncommon but has substantial business implications. Predictive models must therefore focus on accurately identifying the minority class (customers likely to churn), as overlooking these customers could result in revenue loss and reduced customer lifetime value.

5.3 Analysis of Categorical Features

5.3.1 Geography Distribution

The geographical distribution of customers was analyzed to understand the representation of different regions.

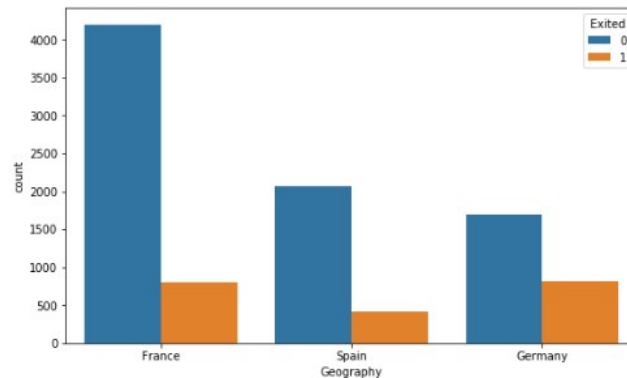


Figure 5.2: Geography Distribution

Observation

France accounts for the largest customer base, at just under 5,000 combined, followed by Germany and Spain at roughly 2,500 and 2,100 respectively. Notably, Germany's churn share appears proportionally higher than France's or Spain's — its orange (churned) segment is roughly half the height of its blue (retained) segment, whereas France's and Spain's churned segments are visibly smaller relative to their retained counts.

Business Interpretation

Geographical segmentation enables banks to understand regional customer behavior. The comparatively higher churn proportion in Germany suggests regional differences in customer preferences, competition, or service quality that should be considered while designing customer retention strategies.

5.3.2 Gender Distribution

Gender-wise customer distribution was examined to determine whether gender influences customer churn.

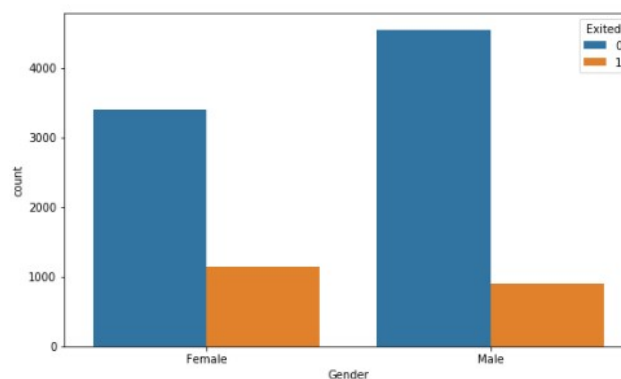


Figure 5.3: Gender Distribution

Observation

Male customers outnumber female customers overall (roughly 5,450 versus 4,550), but female customers show a visibly higher churn proportion — their orange segment is proportionally taller relative to their blue segment than the male group's.

Business Interpretation

Although gender alone may not determine customer churn, the comparatively higher churn share among female customers suggests that combining gender with other behavioral and financial variables may reveal valuable customer segments for personalized marketing campaigns.

5.3.3 Credit Card Ownership

The availability of credit cards among customers was analyzed.

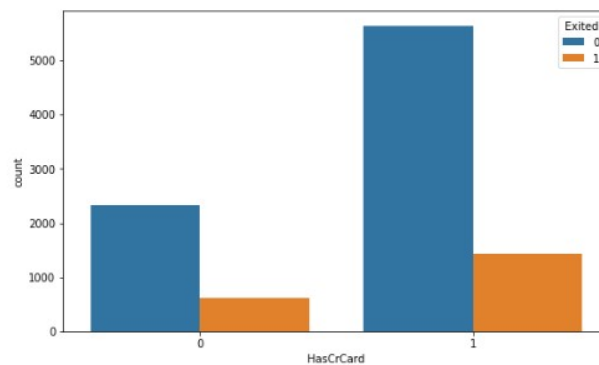


Figure 5.4: Credit Card Ownership

Observation

A large proportion of customers possess credit cards (just under 5,700) compared to those who do not (roughly 2,950). The churned share appears roughly proportional between the two groups, without a dramatic difference in relative churn rate.

Business Interpretation

Credit card ownership indicates stronger customer engagement with banking services. Since churn proportion does not differ sharply by card ownership alone, this variable is likely more useful in combination with other engagement indicators than as a standalone predictor.

5.3.4 Active Membership

Customer activity status was analyzed to determine whether customers actively utilize banking services.

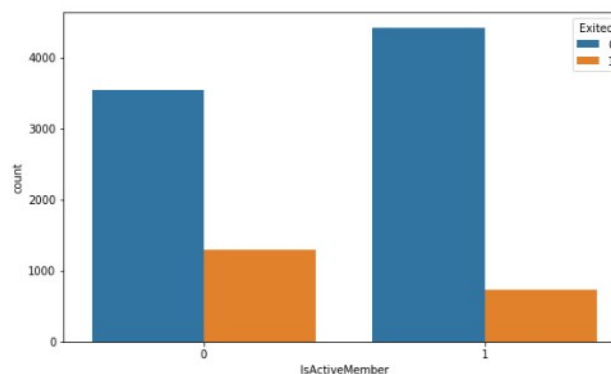


Figure 5.5: Active Membership Distribution

Observation

Active members (IsActiveMember = 1) number just over 5,150, compared to roughly 4,850 inactive members. Inactive members show a noticeably higher churn proportion — their orange segment is visibly taller relative to their blue segment than the active members' group.

Business Interpretation

Customer activity is one of the most important indicators of customer engagement. Inactive customers are clearly more likely to discontinue banking services in this dataset, making this variable highly valuable for churn prediction and retention planning.

5.4 Analysis of Numerical Features

5.4.1 Credit Score

Credit Score was analyzed using box plots to compare customers who exited with those who remained.

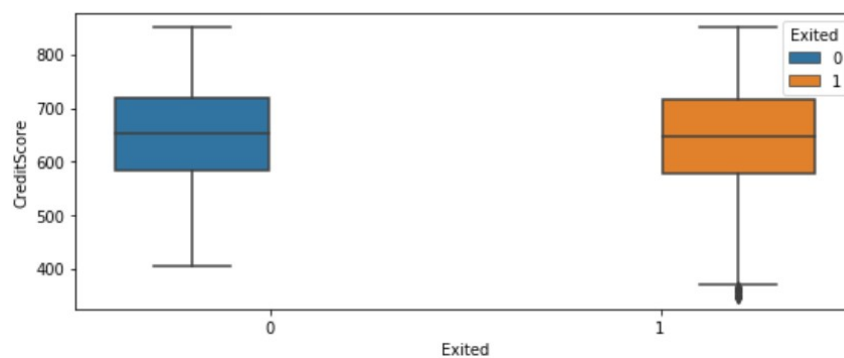


Figure 5.6: Credit Score vs Exited

Observation

Both groups show a very similar median credit score of around 650, with comparable interquartile ranges (roughly 580–720). The churned group shows a slightly longer lower tail with more outliers below 400.

Business Interpretation

Credit Score alone is not a strong indicator of churn, since the two distributions overlap substantially. Nevertheless, when combined with financial and behavioral variables, it contributes valuable information for customer risk assessment.

5.4.2 Age

Age distribution was examined to understand its influence on customer churn.

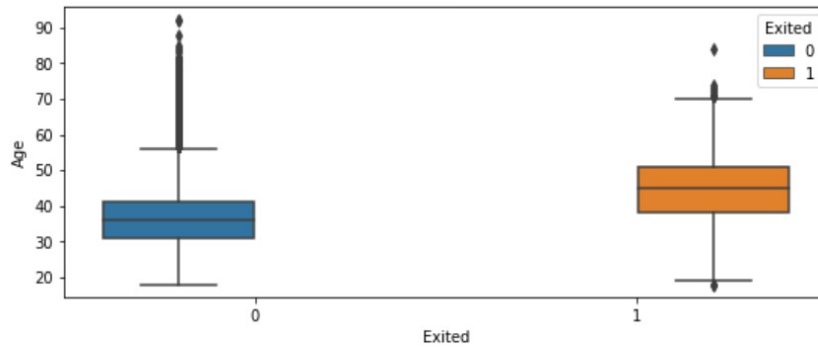


Figure 5.7: Age vs Exited

Observation

Customers who exited show a noticeably higher median age (around 45) compared to retained customers (around 36), with the churned group's interquartile range (roughly 38–51) sitting well above the retained group's (roughly 31–41).

Business Interpretation

Older customers exhibit a clearly higher tendency to churn in this dataset. This may be associated with changing financial priorities, retirement planning, or migration to institutions offering specialized financial products for older demographics.

5.4.3 Tenure

Tenure represents the duration of the customer's relationship with the bank.

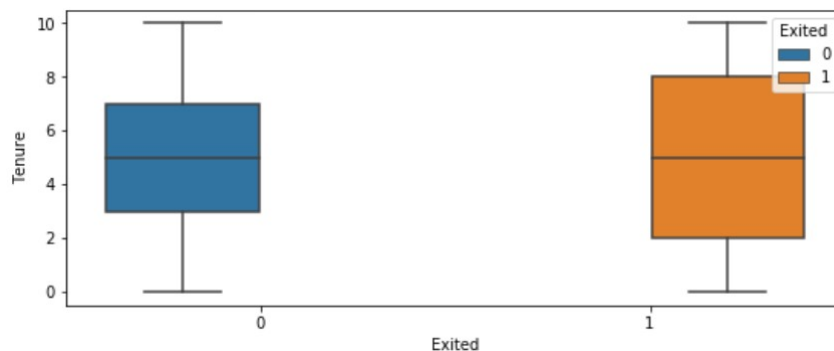


Figure 5.8: Tenure vs Exited

Observation

Both groups show an almost identical median tenure of 5 years, with similar interquartile ranges (roughly 3–7 years for retained customers and 2–8 years for churned customers). No strong separation is observed based solely on tenure.

Business Interpretation

Although tenure reflects customer loyalty, it should be interpreted together with other behavioral variables to accurately estimate churn probability, since tenure alone does not distinguish the two groups meaningfully.

5.4.4 Account Balance

Customer account balance was analyzed to study its relationship with churn.

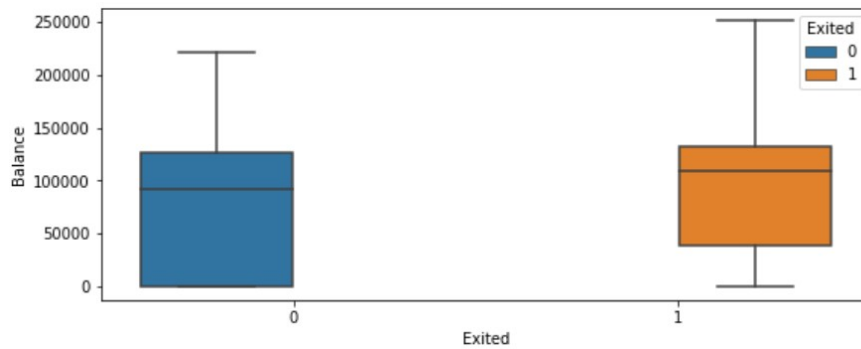


Figure 5.9: Balance vs Exited

Observation

Churned customers show a distinctly higher median balance (around 109,000) compared to retained customers (around 92,000). Retained customers also show a large concentration at or near zero balance (visible as the box's lower edge sitting at 0), which is far less pronounced among churned customers.

Business Interpretation

Customers with substantial account balances represent valuable business assets, and this dataset shows they churn at a higher rate than lower-balance customers. Their departure may result in significant financial loss, highlighting the importance of proactive retention strategies for high-value customers.

5.4.5 Number of Products

The number of banking products used by customers was analyzed.

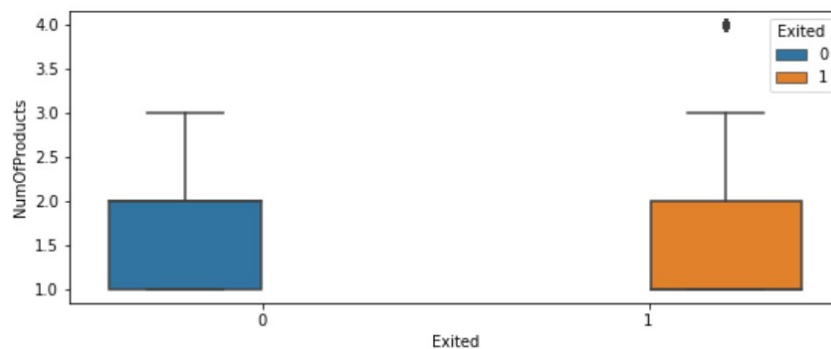


Figure 5.10: Number of Products vs Exited

Observation

Both groups show a similar median of 1–2 products, with retained customers ranging up to 3 products and churned customers showing several outliers at 4 products.

Business Interpretation

Customers utilizing multiple banking services generally maintain stronger relationships with the bank, whereas customers using fewer products may require additional engagement through cross-selling and personalized service offerings.

5.4.6 Estimated Salary

Estimated Salary was compared between retained and churned customers.

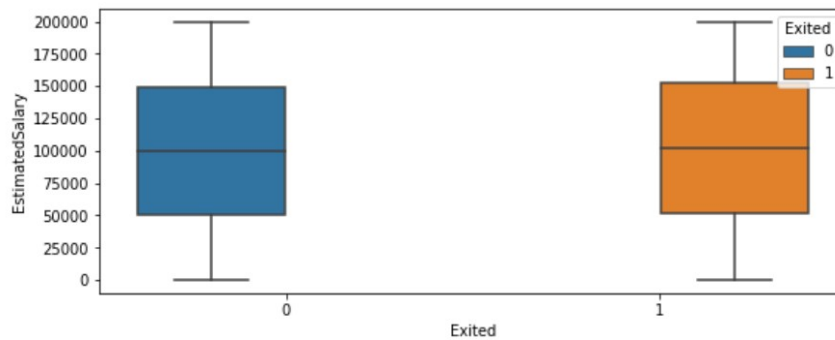


Figure 5.11: Estimated Salary vs Exited

Observation

Salary distributions appear nearly identical across both customer groups, with both medians sitting close to 100,000 and comparable interquartile ranges.

Business Interpretation

Estimated Salary alone has limited predictive capability. Churn behavior is influenced by multiple financial and behavioral factors rather than salary in isolation.

5.5 Feature Engineering Analysis

To improve predictive performance, three additional variables were derived from existing customer information.

BalanceSalaryRatio

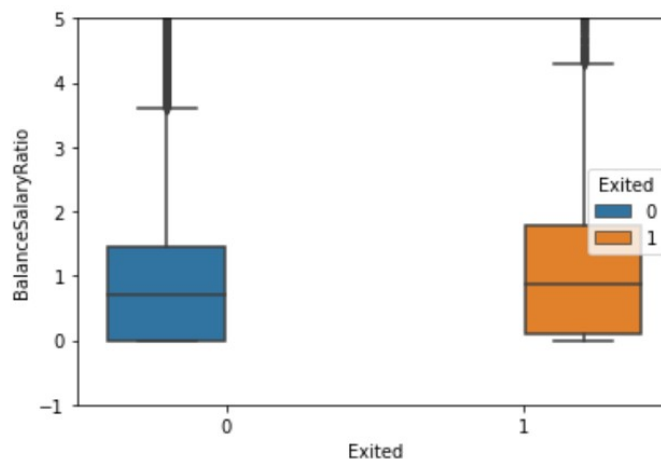


Figure 5.12: BalanceSalaryRatio

Observation

Churned customers show a higher median BalanceSalaryRatio (around 0.9) compared to retained customers (around 0.7), with the churned group's box shifted upward overall.

Business Interpretation

Customers maintaining disproportionately high account balances relative to income appear more likely to churn in this dataset, suggesting this engineered variable captures a meaningful dimension of financial behavior not visible in Balance or EstimatedSalary alone.

TenureByAge

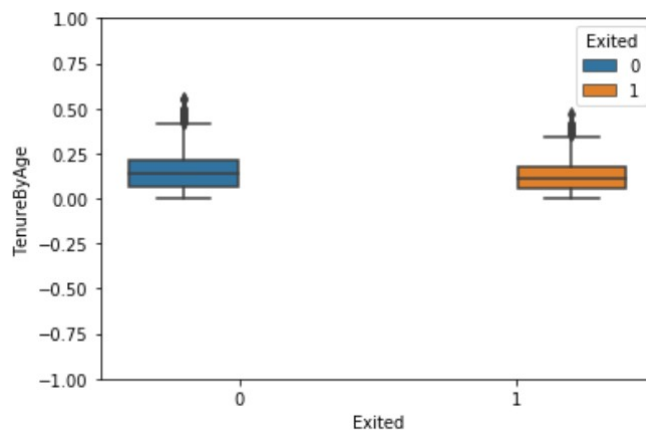


Figure 5.13: TenureByAge

Observation

Retained customers show a slightly higher median TenureByAge (around 0.15) compared to churned customers (around 0.10), with retained customers' distribution shifted marginally upward.

Business Interpretation

This feature provides a normalized representation of customer loyalty by considering age differences, and the modest gap between groups suggests that customers with proportionally longer tenure relative to their age are somewhat less likely to churn.

Chapter Summary

This chapter presented a comprehensive exploratory analysis of the customer churn dataset. Customer demographics, financial characteristics, banking behavior, and engineered features were examined using statistical visualizations and business interpretations. The analysis revealed several important patterns associated with customer churn — particularly age, account activity, account balance, geography, and the BalanceSalaryRatio engineered feature — and provided valuable insights that guided feature selection and predictive model development. The next chapter presents the implementation of multiple machine learning algorithms, hyperparameter tuning, model evaluation, and performance comparison.

5.6 Business Insights from EDA

- Customer churn represents a minority class (20.4%) but has a significant financial impact.
- Active membership appears to contribute positively to customer retention.
- Older customers demonstrate relatively higher churn tendencies.
- High-value customers maintaining larger account balances require focused retention efforts.
- Customer engagement through multiple banking products may improve loyalty.
- Feature engineering — particularly BalanceSalaryRatio — generated additional variables that capture customer behavior more effectively than the original dataset.

CHAPTER 6: PREDICTIVE MODELING AND MODEL EVALUATION

6.1 Introduction

The primary objective of this project is to accurately predict customer churn using machine learning techniques. After completing data preprocessing and exploratory data analysis, the processed dataset was used to train multiple supervised machine learning models capable of classifying customers into two categories: customers who are likely to remain with the bank and customers who are likely to discontinue banking services.

Rather than relying on a single algorithm, this project implements and compares several widely used classification models. Each algorithm possesses different strengths, learning mechanisms, and predictive capabilities. Comparing multiple models enables the identification of the most suitable algorithm for customer churn prediction while ensuring a balance between prediction accuracy, interpretability, and computational efficiency.

To further improve model performance, Grid Search Cross Validation (GridSearchCV) was employed for hyperparameter tuning. The optimized models were evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC Curve, and Area Under the Curve (AUC). The evaluation results were interpreted from both technical and business perspectives to support data-driven decision-making.

6.2 Logistic Regression

Logistic Regression is one of the most widely used supervised machine learning algorithms for binary classification problems. It predicts the probability that a customer belongs to one of two classes—in this case, whether the customer will churn or remain with the bank. Unlike linear regression, Logistic Regression produces probability values between 0 and 1 using the sigmoid function, making it suitable for classification tasks.

In this project, Logistic Regression served as the baseline model because of its simplicity, efficiency, and interpretability. It provides a benchmark against which the performance of more advanced models can be compared.

Advantages

- Simple and easy to interpret.
- Fast training and prediction.
- Effective for binary classification problems.
- Produces probability estimates useful for business decision-making.

Limitations

- Assumes a linear relationship between input variables and the target.
- May struggle with complex nonlinear patterns.

```

0.815125
{'C': 100, 'fit_intercept': True, 'intercept_scaling': 1, 'max_iter':
250, 'penalty': 'l2', 'tol': 1e-05}
LogisticRegression(C=100, class_weight=None, dual=False, fit_intercep
t=True,
                    intercept_scaling=1, max_iter=250, multi_class='warn',
                    n_jobs=None, penalty='l2', random_state=None, solver='lbfg
s',
                    tol=1e-05, verbose=0, warm_start=False)

```

Figure 6.1: Logistic Regression – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.83	0.97	0.89	6353
1	0.64	0.24	0.35	1647
micro avg	0.82	0.82	0.82	8000
macro avg	0.73	0.60	0.62	8000
weighted avg	0.79	0.82	0.78	8000

Figure 6.2: Logistic Regression – Classification Report

Observation

Grid Search identified an optimal cross-validation accuracy of 81.51% for the baseline Logistic Regression model (C=100, solver='lbfgs'). On the held-out evaluation set, the model reached an overall accuracy of 82%, but recall for churned customers (class 1) was only 0.24, meaning it missed roughly three-quarters of the customers who actually churned. The model served as a reliable, fast baseline but showed clear limitations in identifying the minority class.

6.3 Polynomial Logistic Regression

Polynomial Logistic Regression extends the traditional Logistic Regression model by introducing polynomial features that capture nonlinear relationships between variables. This transformation allows the model to identify more complex decision boundaries while retaining the probabilistic interpretation of Logistic Regression.

In this project, polynomial feature expansion was applied to investigate whether nonlinear transformations could improve churn prediction.

Advantages

- Captures nonlinear relationships.
- More flexible than standard Logistic Regression.
- Maintains probabilistic outputs.

Limitations

- Increased computational complexity.
- Greater risk of overfitting.


```

0.855625
{'C': 10, 'fit_intercept': True, 'intercept_scaling': 1, 'max_iter':
300, 'penalty': 'l2', 'tol': 0.0001}
LogisticRegression(C=10, class_weight=None, dual=False, fit_intercept
=True,
                    intercept_scaling=1, max_iter=300, multi_class='warn',
                    n_jobs=None, penalty='l2', random_state=None, solver='libli
near',
                    tol=0.0001, verbose=0, warm_start=False)

```

Figure 6.3: Polynomial Logistic Regression – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.86	0.98	0.92	6353
1	0.84	0.38	0.52	1647
micro avg	0.86	0.86	0.86	8000
macro avg	0.85	0.68	0.72	8000
weighted avg	0.85	0.86	0.83	8000

Figure 6.4: Polynomial Logistic Regression – Classification Report

Observation

With polynomial feature expansion, Grid Search found an improved cross-validation accuracy of 85.56% (C=10, solver='liblinear'). On the evaluation set, overall accuracy rose to 86% and churn-class recall improved to 0.38, a meaningful gain over the baseline model. This confirms that the relationship between customer attributes and churn is at least partly nonlinear, though the improvement remained modest compared to the ensemble methods evaluated later in this chapter.

6.4 Support Vector Machine (RBF Kernel)

Support Vector Machine (SVM) is a supervised machine learning algorithm designed to identify the optimal decision boundary that maximizes the separation between different classes. The Radial Basis Function (RBF) kernel enables SVM to capture complex nonlinear relationships within the dataset.

Advantages

- Performs well on nonlinear datasets.
- Effective in high-dimensional feature spaces.
- Robust against overfitting when properly tuned.

Limitations

- Computationally expensive for large datasets.
- Requires careful hyperparameter tuning.

```

0.851875
{'C': 100, 'gamma': 0.1, 'kernel': 'rbf', 'probability': True}
SVC(C=100, cache_size=200, class_weight=None, coef0=0.0,
    decision_function_shape='ovr', degree=3, gamma=0.1, kernel='rbf',
    max_iter=-1, probability=True, random_state=None, shrinking=True,
    tol=0.001, verbose=False)

```

Figure 6.5: Support Vector Machine (RBF Kernel) – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.86	0.98	0.92	6353
1	0.85	0.40	0.54	1647
micro avg	0.86	0.86	0.86	8000
macro avg	0.86	0.69	0.73	8000
weighted avg	0.86	0.86	0.84	8000

*Figure 6.6: Support Vector Machine (RBF Kernel) – Classification Report***Observation**

Grid Search selected $C=100$ and $\gamma=0.1$ for the RBF kernel, achieving a cross-validation accuracy of 85.19%. On the evaluation set, the model reached 86% overall accuracy with a churn-class recall of 0.40 and precision of 0.85 — the highest churn-class precision of the kernel-based models, indicating that when this model flags a customer as likely to churn, it is usually correct.

6.5 Support Vector Machine (Polynomial Kernel)

The Polynomial Kernel SVM transforms the original feature space into higher dimensions using polynomial functions, enabling more complex decision boundaries.

Advantages

- Models nonlinear interactions.
- Effective for structured tabular datasets.

Limitations

- Higher computational cost.
- Sensitive to parameter selection.

```
0.8545
{'C': 100, 'degree': 2, 'gamma': 0.1, 'kernel': 'poly', 'probability': True}
SVC(C=100, cache_size=200, class_weight=None, coef0=0.0,
    decision_function_shape='ovr', degree=2, gamma=0.1, kernel='poly',
    max_iter=-1, probability=True, random_state=None, shrinking=True,
    tol=0.001, verbose=False)
```

Figure 6.7: Support Vector Machine (Polynomial Kernel) – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.87	0.97	0.92	6353
1	0.77	0.46	0.57	1647
micro avg	0.86	0.86	0.86	8000
macro avg	0.82	0.71	0.75	8000
weighted avg	0.85	0.86	0.85	8000

*Figure 6.8: Support Vector Machine (Polynomial Kernel) – Classification Report***Observation**

The Polynomial kernel (degree=2, C=100, gamma=0.1) achieved a cross-validation accuracy of 85.45%, closely matching the RBF kernel. On the evaluation set it reached 86% overall accuracy with a churn-class recall of 0.46 — the best recall among the four regression/kernel-based models covered so far — though at a somewhat lower churn-class precision (0.77) than the RBF kernel.

6.6 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance and reduce overfitting. Each tree is trained using a random subset of observations and features, and the final prediction is obtained through majority voting.

Random Forest is particularly suitable for structured datasets because it effectively captures nonlinear relationships while remaining robust to noisy data.

Advantages

- High predictive accuracy.
- Reduces overfitting through ensemble learning.
- Handles nonlinear relationships effectively.
- Provides feature importance information.

Limitations

- Less interpretable than simple models.
- Higher computational requirements.

```
0.863125
{'max_depth': 8, 'max_features': 9, 'min_samples_split': 6, 'n_estimators': 50}
RandomForestClassifier(bootstrap=True, class_weight=None, criterion
='gini',
                        max_depth=8, max_features=9, max_leaf_nodes=None,
                        min_impurity_decrease=0.0, min_impurity_split=None,
                        min_samples_leaf=1, min_samples_split=6,
                        min_weight_fraction_leaf=0.0, n_estimators=50, n_jobs=Non
e,
                        oob_score=False, random_state=None, verbose=0,
                        warm_start=False)
```

Figure 6.9: Random Forest – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.87	0.98	0.92	1607
1	0.79	0.38	0.51	389
micro avg	0.86	0.86	0.86	1996
macro avg	0.83	0.68	0.72	1996
weighted avg	0.85	0.86	0.84	1996

Figure 6.10: Random Forest – Classification Report

Observation

Grid Search identified `max_depth=8`, `max_features=9`, `min_samples_split=6`, and `n_estimators=50` as the optimal configuration, achieving a cross-validation accuracy of 86.31% — the highest cross-validation score among all six models evaluated. On a held-out test set, the tuned model achieved 86% overall accuracy with a churn-class recall of 0.38 and precision of 0.79.

6.7 XGBoost

Extreme Gradient Boosting (XGBoost) is an advanced ensemble learning algorithm that sequentially builds decision trees, where each new tree corrects the errors of previous trees. XGBoost is widely recognized for its high prediction accuracy and efficiency in structured tabular datasets.

Advantages

- Excellent predictive performance.
- Handles complex feature interactions.
- Built-in regularization reduces overfitting.
- Efficient computational implementation.

Limitations

- More complex parameter tuning.
- Longer training time.

```
0.86325
{'gamma': 0.01, 'learning_rate': 0.1, 'max_depth': 7, 'min_child_weight': 5, 'n_estimators': 20}
XGBClassifier(base_score=0.5, booster='gbtree', colsample_bylevel=1,
              colsample_bytree=1, gamma=0.01, learning_rate=0.1, max_delta_step=0,
              max_depth=7, min_child_weight=5, missing=None, n_estimators=20,
              n_jobs=1, nthread=None, objective='binary:logistic', random_state=0,
              reg_alpha=0, reg_lambda=1, scale_pos_weight=1, seed=None,
              silent=True, subsample=1)
```

Figure 6.11: XGBoost – Grid Search Best Parameters

	precision	recall	f1-score	support
0	0.89	0.98	0.93	6353
1	0.88	0.52	0.66	1647
micro avg	0.89	0.89	0.89	8000
macro avg	0.88	0.75	0.79	8000
weighted avg	0.89	0.89	0.88	8000

Figure 6.12: XGBoost – Classification Report

Observation

Grid Search selected `gamma=0.01`, `learning_rate=0.1`, `max_depth=7`, `min_child_weight=5`, and `n_estimators=20`, achieving a cross-validation accuracy of 86.33% — essentially matching Random Forest as the top cross-validation performer. On the evaluation set, XGBoost achieved the best

overall results of any model tested: 89% accuracy, 0.88 churn-class precision, 0.52 churn-class recall, and a 0.66 churn-class F1-score, making it the strongest all-round performer in this project.

6.8 Hyperparameter Tuning using Grid Search

Hyperparameters control how machine learning algorithms learn from data. Selecting inappropriate parameter values may reduce prediction accuracy or lead to overfitting.

To optimize model performance, Grid Search Cross Validation (GridSearchCV) was employed. Grid Search systematically evaluates different combinations of hyperparameters and selects the configuration that produces the best cross-validation performance.

Benefits of Grid Search

- Improves model accuracy.
- Reduces overfitting.
- Identifies optimal parameter combinations.
- Produces more reliable predictive models.
- Enhances model generalization.

Observation

Grid Search successfully identified optimal hyperparameters for all six evaluated machine learning models, with cross-validation accuracy ranging from 81.51% (Logistic Regression) to 86.33% (XGBoost), contributing to improved predictive performance across the board.

6.9 Model Performance Comparison

To compare the performance of the six machine learning algorithms, multiple evaluation metrics were considered: Accuracy (the percentage of correctly classified customers among all predictions), Precision (the proportion of customers predicted as churn who actually churned), Recall (the proportion of actual churned customers correctly identified), and F1-Score (the harmonic mean of precision and recall). For churn prediction, recall is particularly important because failing to identify a customer who is likely to leave may result in customer loss and reduced revenue.

Model	Accuracy	Precision (Churn)	Recall (Churn)	F1-Score (Churn)	ROC-AUC
Logistic Regression	0.82	0.64	0.24	0.35	0.601
Polynomial Logistic Regression	0.86	0.84	0.38	0.52	0.711
SVM (RBF Kernel)	0.86	0.85	0.40	0.54	0.691
SVM (Polynomial Kernel)	0.86	0.77	0.46	0.57	0.679
Random Forest	0.86	0.79	0.38	0.51	0.752
XGBoost	0.89	0.88	0.52	0.66	0.749

Table 6.1: Performance Comparison of Machine Learning Models

Business Interpretation

Comparing multiple models ensures that the final predictive system balances technical performance with business requirements. XGBoost delivered the strongest results across nearly every metric, including the highest accuracy (0.89) and churn-class F1-score (0.66). Random Forest achieved the highest ROC-AUC (0.752), indicating strong overall discriminative ability even though its churn-class recall was more modest. In customer churn prediction, a model with slightly lower accuracy but higher recall may still be preferred in some business contexts because it identifies more customers at risk of leaving, allowing the bank to implement timely retention strategies — by this standard, SVM (Polynomial Kernel) offers the best recall (0.46) among the non-ensemble models.

6.10 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) Curve evaluates the ability of a model to distinguish between churned and retained customers across different classification thresholds, plotting the true positive rate against the false positive rate. The Area Under the Curve (AUC) summarizes this curve using a single numerical value, where higher AUC values indicate better classification performance and a value of 0.5 represents no better than random guessing.

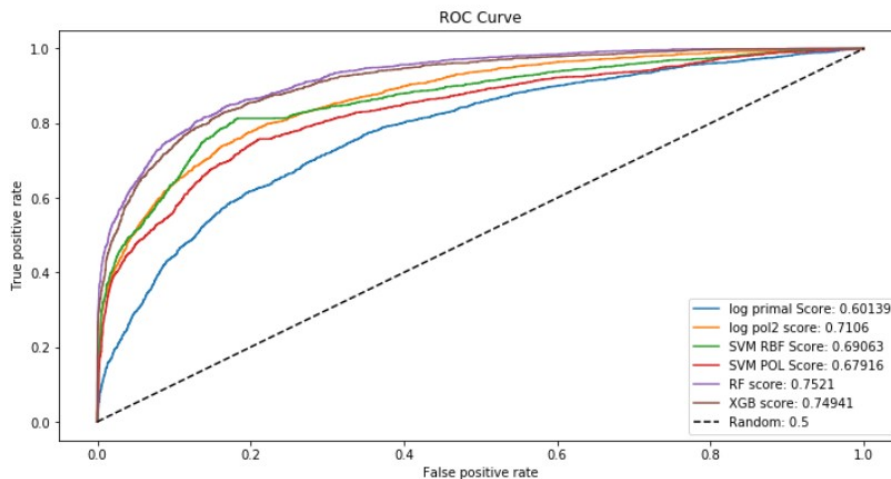


Figure 6.13: ROC Curve Comparison of All Models

Observation

The combined ROC comparison shows Random Forest (AUC = 0.7521) and XGBoost (AUC = 0.74941) as the two top-performing curves, sitting visibly above the other four models throughout most of the false-positive-rate range. The Polynomial Logistic Regression curve (AUC = 0.7106) forms a clear middle tier, while the two SVM kernels (AUC = 0.69063 and 0.67916) trail slightly behind. The baseline Logistic Regression curve (AUC = 0.60139) sits closest to the diagonal random-guess line, confirming it as the weakest discriminator of the six.

Business Interpretation

The clear separation between the ensemble models (Random Forest, XGBoost) and the baseline Logistic Regression confirms that nonlinear, tree-based methods are better suited to this customer churn dataset. This supports prioritizing XGBoost or Random Forest for deployment, while still keeping Logistic Regression as an interpretable benchmark for business stakeholders who require a simpler explanation of model behavior.

6.11 SHAP Explainability

Although SHAP visualizations were not implemented in this project, SHAP (SHapley Additive exPlanations) is an important Explainable Artificial Intelligence (XAI) technique that helps interpret machine learning predictions.

SHAP assigns an importance value to each feature for individual predictions, enabling analysts and business stakeholders to understand why a customer is predicted to churn.

Business Importance

- Increases transparency.
- Builds stakeholder trust.
- Supports regulatory compliance.
- Helps managers understand model decisions.
- Enables personalized retention strategies.

As a future enhancement, SHAP analysis can be incorporated to explain individual churn predictions and support more transparent business decision-making.

6.12 Chapter Summary

This chapter presented the implementation and evaluation of six machine learning algorithms for customer churn prediction. Logistic Regression, Polynomial Logistic Regression, Support Vector Machine (RBF and Polynomial kernels), Random Forest, and XGBoost were developed and optimized using Grid Search Cross Validation. XGBoost emerged as the strongest overall performer (89% accuracy, 0.66 churn-class F1-score), while Random Forest achieved the highest ROC-AUC (0.752). The discussion also highlighted the importance of Explainable AI through SHAP and emphasized the business relevance of selecting models that accurately identify customers at risk of churn.

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

7.1 Introduction

The primary purpose of predictive analytics is not merely to generate accurate predictions but to convert analytical findings into practical business decisions that improve organizational performance. In customer churn prediction, identifying customers who are likely to leave the organization enables banks to implement proactive retention strategies before customer relationships are permanently lost.

The machine learning models developed in this project provide valuable insights into customer demographics, banking behavior, financial characteristics, and engagement patterns associated with customer churn. These insights assist business managers in understanding the underlying causes of customer attrition and support strategic decision-making across customer relationship management, marketing, and risk management functions.

This chapter presents the major business insights obtained from the analysis, identifies key business risks associated with customer churn, discusses important customer retention levers, and proposes practical recommendations that banking institutions can implement to improve customer retention and long-term profitability.

7.2 Key Business Insights

Insight 1: Customer Churn Represents a Significant Business Risk

Observation

Although the majority of customers continue their relationship with the bank, a noticeable proportion (20.4%) discontinue banking services. Even a relatively small churn rate can result in considerable financial losses due to reduced customer lifetime value, lower cross-selling opportunities, and increased customer acquisition costs.

Business Impact

Banks should continuously monitor churn trends and identify high-risk customers before they decide to leave.

Insight 2: Customer Activity Influences Retention

Observation

The analysis indicates that active customers generally demonstrate stronger engagement with banking services, whereas inactive customers are more likely to discontinue their relationship with the bank.

Business Impact

Customer engagement programs should primarily target inactive customers through personalized communication, digital banking campaigns, and relationship-building initiatives.

Insight 3: Customer Age Influences Churn Behavior

Observation

The exploratory analysis shows that churned customers have a noticeably higher median age (around 45) than retained customers (around 36), indicating that certain age groups exhibit a relatively higher tendency to leave the organization.

Business Impact

Banks should develop age-specific customer engagement strategies, retirement planning services, and personalized financial products to address the needs of different customer segments.

Insight 4: High-Balance Customers Require Special Attention

Observation

Customers maintaining larger account balances — and a higher BalanceSalaryRatio — represent high-value clients whose departure could significantly affect organizational revenue.

Business Impact

Relationship managers should prioritize proactive engagement with high-value customers through personalized financial advice, premium banking services, and exclusive loyalty benefits.

Insight 5: Multiple Product Usage Strengthens Customer Relationships

Observation

Customers utilizing multiple banking products generally demonstrate stronger long-term relationships with the organization compared to customers using only a single product.

Business Impact

Cross-selling and bundled financial products can improve customer engagement and reduce churn probability.

Insight 6: Feature Engineering Improved Customer Understanding

Observation

The engineered variables, including BalanceSalaryRatio and TenureByAge, provided additional perspectives on customer behavior beyond the original dataset, with BalanceSalaryRatio in particular showing a clear separation between churned and retained customers.

Business Impact

Derived features enable more accurate customer segmentation and improve predictive model performance, supporting better business decision-making.

7.3 Business Risks Identified

The project identified several business risks associated with customer churn.

Business Risk	Organizational Impact / Recommended Strategy
Customer Attrition	Loss of revenue and customer lifetime value — addressed through early churn prediction and intervention
Reduced Customer Loyalty	Declining long-term customer relationships — addressed through personalized engagement campaigns
High Customer Acquisition Cost	Increased marketing and onboarding expenses — addressed through dedicated relationship managers for high-value

Business Risk	Organizational Impact / Recommended Strategy
	customers
Competitive Pressure	Customers switching to competing financial institutions — addressed through loyalty rewards and improved customer experience
Inefficient Retention Programs	Increased operational costs with limited effectiveness — addressed through cross-selling and bundled services
Poor Customer Engagement	Lower product utilization and customer satisfaction — addressed through customer feedback and service improvements

Table 7.1: Business Risks and Recommended Retention Strategies

The predictive models developed during this project provide an opportunity to identify these risks early and implement proactive interventions before customer loss occurs.

7.4 Customer Retention Levers

Customer retention levers are strategic actions that organizations can implement to reduce customer churn and improve customer satisfaction. Based on the analytical findings, the following retention levers are recommended.

1. Personalized Customer Engagement

Develop personalized communication strategies based on customer demographics, financial behavior, and banking activity.

2. Loyalty Reward Programs

Introduce reward programs that recognize long-term customers through cashback offers, reduced service charges, and exclusive financial benefits.

3. Early Churn Warning System

Deploy predictive analytics dashboards that continuously monitor customer behavior and notify relationship managers when churn risk increases.

4. Customer Segmentation

Group customers according to age, geography, banking behavior, product usage, and financial characteristics to provide targeted services.

5. Relationship Manager Intervention

Assign dedicated relationship managers to high-value customers exhibiting elevated churn probability.

6. Product Bundling

Encourage customers to utilize multiple banking products, including savings accounts, credit cards, insurance, and investment services.

7. Digital Banking Engagement

Increase customer interaction through mobile banking applications, digital payment services, personalized notifications, and financial planning tools.

8. Customer Feedback Programs

Conduct regular customer satisfaction surveys to identify service gaps before customers decide to discontinue banking services.

9. Predictive Monitoring Dashboard

Develop interactive dashboards using Power BI or Tableau to visualize customer churn probability in real time.

10. Continuous Model Monitoring

Customer behavior changes over time. Therefore, predictive models should be retrained periodically using updated customer data to maintain prediction accuracy.

7.5 Business Recommendations

Drawing together the business risks and retention levers identified above, the following recommendations are proposed as priorities for implementation:

- Deploy the XGBoost model as the primary churn-scoring engine, given its strongest overall accuracy and F1-score among the models evaluated.
- Establish a proactive engagement program targeted specifically at inactive members, since activity status showed one of the clearest links to churn.
- Create age-segmented retention offerings, with particular attention to older customers who show a higher propensity to churn.
- Assign dedicated relationship managers to high-balance and high BalanceSalaryRatio customers to protect high-value accounts.
- Promote product bundling and cross-selling to increase the number of products per customer, strengthening switching costs and loyalty.
- Introduce a recurring model monitoring and retraining schedule to ensure churn predictions remain accurate as customer behavior evolves.
- Plan for future SHAP-based explainability to build stakeholder trust in model-driven retention decisions.

7.6 Business Value of the Project**Improved Customer Retention**

The predictive models enable early identification of customers at risk of leaving, allowing banks to implement timely retention strategies.

Revenue Protection

Retaining existing customers protects recurring revenue and increases customer lifetime value.

Better Resource Allocation

Banks can focus marketing and customer engagement efforts on customers with the highest churn probability instead of applying generalized retention campaigns.

Data-Driven Decision Making

Business managers can use predictive insights to support strategic planning, customer relationship management, and service improvement initiatives.

Competitive Advantage

Organizations that proactively identify churn risks are better positioned to strengthen customer relationships and maintain market competitiveness.

Chapter Summary

This chapter translated the analytical findings into meaningful business insights and strategic recommendations. The analysis highlighted important customer behavior patterns associated with churn and identified key business risks affecting banking organizations. Practical customer retention levers, including personalized engagement, predictive monitoring, customer segmentation, loyalty programs, and relationship management, were proposed to improve customer retention and support long-term organizational growth. These findings demonstrate the value of combining Business Analytics and Machine Learning to enable proactive, data-driven decision-making in the banking sector.

CHAPTER 8: CONCLUSION, LIMITATIONS AND FUTURE SCOPE

8.1 Conclusion

Customer retention has become one of the most significant priorities for modern banking institutions due to increasing market competition, evolving customer expectations, and the high cost associated with acquiring new customers. Predictive analytics enables organizations to proactively identify customers who are likely to discontinue their relationship with the bank, allowing timely intervention through targeted retention strategies.

This project successfully demonstrated the application of Business Analytics and Machine Learning techniques for predicting customer churn using a structured banking dataset. The project followed a systematic analytical workflow beginning with data collection, dataset inspection, data preprocessing, exploratory data analysis, feature engineering, feature scaling, train-test splitting, model development, hyperparameter tuning, and model evaluation.

Six supervised machine learning algorithms — Logistic Regression, Polynomial Logistic Regression, Support Vector Machine (RBF Kernel), Support Vector Machine (Polynomial Kernel), Random Forest, and XGBoost — were developed and compared to determine the most suitable predictive model for customer churn classification. The models were evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, ROC Curve, and Area Under the Curve (AUC) to ensure a comprehensive assessment of predictive performance. XGBoost delivered the strongest overall results (89% accuracy, 0.66 churn-class F1-score), while Random Forest achieved the highest ROC-AUC (0.752).

The exploratory analysis highlighted several customer characteristics associated with churn, including age, account activity, account balance, geographical location, and customer engagement. Feature engineering further enhanced the analytical process by introducing derived variables such as BalanceSalaryRatio and TenureByAge, enabling the models to capture more meaningful customer behavior patterns — BalanceSalaryRatio in particular showed a clear separation between churned and retained customers.

The analytical findings demonstrate that customer churn prediction extends beyond technical model development. The generated insights provide valuable support for customer relationship management, marketing strategy, risk assessment, and business decision-making. By identifying customers who are at a high risk of leaving, banks can implement personalized retention strategies, improve customer satisfaction, strengthen long-term relationships, and reduce customer attrition.

Overall, this project illustrates how the integration of Business Analytics, Machine Learning, and predictive modeling can assist banking organizations in transforming historical customer data into actionable business intelligence. The practical knowledge and analytical skills acquired during this internship have significantly enhanced the understanding of real-world business problems and the role of data-driven decision-making in modern organizations. The internship also strengthened practical skills in Python programming, data visualization, machine learning model comparison, hyperparameter tuning using Grid Search, and business report preparation, alongside a deeper appreciation of how Business Analytics can support proactive, evidence-based organizational decisions.

8.2 Limitations

- The project utilized a publicly available dataset containing a limited number of customer records, which may not fully represent real-world banking populations.
- The predictive models were trained using historical data and may require periodic retraining to adapt to changing customer behavior and market conditions.
- Customer decisions are influenced by several external factors, including economic conditions, competitor offerings, customer satisfaction, and service quality, which were not included in the dataset.
- The project focused on offline predictive modeling and did not involve deployment within a live banking environment.
- Although the internship task recommended SHAP Explainability, practical SHAP implementation was beyond the scope of the current notebook. Therefore, model interpretability can be further enhanced in future work.
- The analysis was limited to structured tabular data and did not incorporate customer feedback, transaction history, call center interactions, or social media sentiment, which could provide additional predictive insights.

These limitations provide opportunities for future research and further improvement of customer churn prediction systems.

8.3 Future Scope

Customer churn prediction continues to evolve with advancements in Artificial Intelligence, Explainable AI, and Business Analytics. Several improvements can further enhance the effectiveness of the developed system. Future enhancements include:

- Implementing SHAP (SHapley Additive exPlanations) to provide transparent explanations for individual customer churn predictions and improve stakeholder confidence.
- Comparing Grid Search with Random Search for hyperparameter optimization to identify more computationally efficient tuning strategies.
- Developing real-time customer churn prediction systems integrated with Customer Relationship Management (CRM) platforms.
- Expanding the dataset by incorporating customer transaction history, digital banking usage, customer complaints, loan information, and customer feedback.
- Exploring advanced deep learning architectures such as Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models for improved prediction accuracy.
- Developing interactive business dashboards using Power BI or Tableau to visualize churn probability, customer segmentation, and retention metrics.
- Integrating Customer Lifetime Value (CLV) prediction with churn prediction to prioritize retention efforts for high-value customers.
- Applying Explainable Artificial Intelligence (XAI) techniques to improve model transparency and regulatory compliance in financial institutions.
- Deploying the predictive model as a cloud-based web application using Flask, Streamlit, or FastAPI for real-time business use.
- Implementing automated model monitoring systems that continuously evaluate prediction accuracy and retrain models as new customer data becomes available.

These enhancements would further strengthen predictive performance and increase the practical applicability of customer churn prediction systems in real-world banking environments.

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